

Dynamic, Static, or None? Virtual Magnification Loupe Designs for High Precision Tool Manipulation: A Dental Case Study

Category: Research

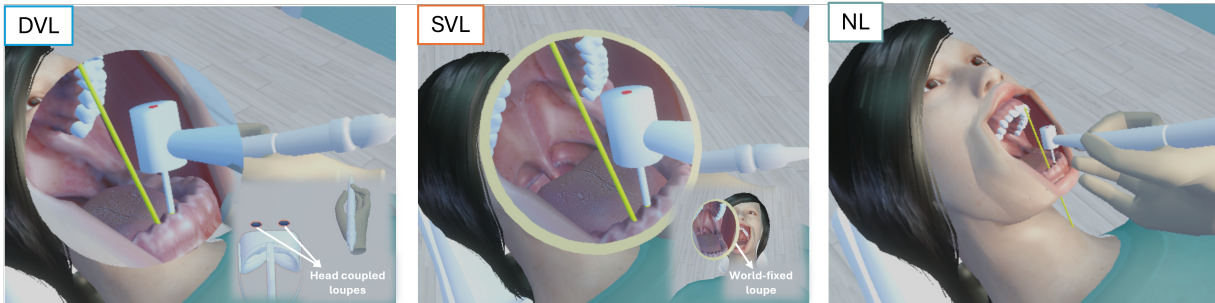


Figure 1: **Co-designed Virtual Magnification Loupe framework:** Dynamic Virtual Loupe (DVL): follows the user's head direction, Static Virtual Loupe (SVL) positioned adjacent to the patient's mouth, No Loupe baseline (NL) without magnification loupe.

ABSTRACT

Precise tool manipulation with magnification loupes is fundamental in dentistry, yet the design of virtual magnification in immersive environments remains underexplored. This paper presents two Virtual Magnification Loupes (VMLs) co-designed with two senior dentists over six months: a Dynamic Virtual Loupe (DVL) providing head-coupled stereoscopic magnification and a Static Virtual Loupe (SVL) providing world-fixed magnification anchored near the intra-oral cavity. Both were integrated in a realistic immersive dental training environment featuring a physical dental treatment room with a phantom-based patient replica, a drill tool, and foot pedal-task confirmation to preserve the clinical workflow. We conducted a within-subjects study ($N = 25$) in which dental students performed high-precision drill positioning tasks under three conditions: DVL, SVL, and a No Loupe (NL) baseline. We measured positional error, rotational error, task completion time, cognitive load (NASA-TLX), and user preference. Results showed no significant differences in completion time or spatial error across conditions. However both VMLs significantly reduced perceived cognitive workload compared to NL. DVL yielded the lowest temporal demand and highest perceived performance, while SVL produced the lowest physical demand. In preference rankings; DVL was preferred by 15 dental students, SVL by nine, and NL by only one. These results indicate that VMLs can achieve equivalent task precision while significantly reducing cognitive load. The complementary strengths of both VML designs suggest that future immersive systems should support switchable loupe modes to accommodate varying task demands.

Index Terms: Magnification loupe, co-design, tool manipulation, mixed reality, dentistry.

1 INTRODUCTION

High-precision visual and motor control is a fundamental requirement in many industrial and professional domains including microelectronics, watchmaking, jewelry fabrication, and microsurgery [64, 58, 60, 40]. In each of these fields, practitioners have to manipulate tools in constrained workspaces, often under challenging lighting conditions and with limited depth cues. The ability to perceive fine spatial details directly determines the task accuracy, operator safety, and outcome quality [65, 9, 18, 35].

A particularly challenging example of these issues appears in dental practice [10, 55, 62], both during daily clinical proce-

dures [25] as well as in training environments [44, 53]. Dental practitioners have to perform fine tool manipulations within the patient's intraoral cavity; a workspace that is highly constrained, poorly illuminated, and partially obscured by soft tissue and instruments [29, 28]. Beyond motor demands, dental procedures require concurrent clinical decision-making, such as determining optimal drill axis and depth, assessing bone density and tissue condition to regulate drilling speed and pressure, and selecting sequential instruments, all while maintaining precise spatial control of the hand-piece [19, 59]. These conditions place particular strain on depth perception and posture maintenance, and are especially challenging for novice practitioners, whose performance has been significantly associated with higher perceived cognitive load during complex dental tasks [30, 2].

Across domains requiring high manual precision, physical magnification loupes (MLs) serve as the primary optical aid for bridging the gap between human visual acuity and the fine scale of manipulation required, enlarging the apparent size of the working field and supporting improved working posture [17, 7, 8, 46]. In modern dental practice, loupes are routinely used across nearly all clinical procedures, ranging from basic diagnostics to surgical interventions such as implant placement [43, 22]. However, their benefits are accompanied by a steep learning curve [1] and high acquisition costs, limiting accessibility in educational institutions and low-resource settings [4, 52].

In parallel, recent advances in augmented and mixed reality (AR/MR) head-mounted displays (HMDs) have enabled the integration of digital interfaces within dental workflows [3, 33, 15]. These systems do not inherently incorporate optical magnifying loupes. However, their programmable control over viewpoint, scale, and overlay rendering positions them as a viable platform for implementing virtual magnification loupes (VMLs), software-driven systems designed to emulate and extend the functional capabilities of conventional MLs. [42, 54, 49, 26].

However, prior work remains limited: existing implementations often rely on rigid static overlays [12, 11] and present a narrow range of VML design alternatives with limited variation in key parameters such as reference-frame coupling, optical placement, or field of view [47, 50]. Furthermore, these systems demonstrate limited adaptation to the user context [56], constrained resolution, and usability issues arising from control schemes, with little comparative evaluation in realistic, domain-specific environments [27, 13]. As a result, key research questions remain open, including how dif-

Table 1: Summary of related work on virtual magnification loupes in AR/VR.

Work	Year	Technology	Design	User Study	Task
Martin-Gonzalez et al. [39]	2009	VST HMD; stereo cameras; gaze-contingent	Linear & hybrid lens (central radial + constrained periphery); world-locked; gaze-activated	$N = 5$ surgeons; within-subjects	Operative task; performance over time
Kluge et al. [32]	2019	VR (Oculus Rift); 6DoF hands	Circular disc lens; world-locked; grabbable 6DoF; resizable; multi-lens overlap; remote proxy	$N > 100$; no controlled exp.	Immersive analytics; subjective only
Qian et al. [47]	2022	OST HMD (Magic Leap One) + Galilean loupe	Hybrid optical+digital: magnified inner zone + AR annular ring; head-locked	$N = 9$ CS students; single condition	AR guidance targeting; sub-mm accuracy
Ilo et al. [27]	2021	VST HMD (Varjo XR-1); eye tracking	Full-FOV zoom (1x-4x); hands-free; dwell menu or wink slider; head-locked	$N = 6$ surgeons & residents; feasibility	Suturing sim.; completion & usability
Schubert et al. [51]	2023	AR HMD (HoloLens 2); camera rendering	Rectangular window; world-locked; uni/bimanual; 5 scale methods (size, distance, voice, device, slider)	$N = 20$; within-subjects	Unspecified; subjective & qualitative only
Kern et al. [31]	2023	VR; virtual camera; surface shader	Planar mirror surface; world-locked; virtual camera crops (3x) + interpolates onto mirror plane; controller-toggled	$N = 14$; within-subjects	Rod alignment; rotational accuracy & distance
Zeng et al. [67]	2024	VST HMD (Varjo XR-1/XR-3); eye tracking	Full-FOV gaze zoom; head-locked; discrete menu or wink control	$N = 6$; two-phase study	Microsurgical suturing; completion & SUS

ferent VML design configurations affect task precision and cognitive workload.

To address this gap this study proposes and evaluates two VMLs interaction designs against a No loupe (NL) baseline integrated into a realistic immersive dental training environment built around a dental treatment room with a phantom-based patient replica and foot pedal activation to preserve the natural hand-based clinical workflow.

The contributions of this work are as follows:

- **Design Analysis:** A problem-driven characterization of the core challenges of deploying virtual magnification in immersive dental simulation, grounded in clinical observation and expert consultation.
- **System Design:** Two fully realized VML systems co-designed with two expert dentists over six months, each representing a distinct magnification paradigm informed by clinical clinical practice: a **Dynamic Virtual Loupe (DVL)**, a head-coupled magnification system whose viewport moves naturally with the practitioner, replicating the intuitive behaviour of optical ML; and a **Static Virtual Loupe (SVL)**, a world-fixed magnification overlay spatially anchored at a procedurally relevant location, approximating the experience of a mounted optical aid.
- **User Evaluation:** A controlled within-subjects study ($N = 25$) with dental students performing precision tool-positioning tasks across DVL, SVL, and a NL baseline (see Fig. 1), providing empirical evidence on the effects of VML interaction design on task performance (positional error, rotational error, completion time), cognitive load (NASA-TLX), and user preference.

2 RELATED WORKS

This section reviews related work on virtual magnification systems (see Tab. 1) chronologically, focusing on loupe design, interaction paradigms, and empirical evidence of performance benefit, to identify the key limitations that motivate the present contribution.

Martin-Gonzalez et al. [39] introduced two VML magnification designs (a custom-built HMD equipped with two color stereo cameras and a hybrid dual-tracking setup): a linear design analogous to

conventional optical loupes, and a hybrid design combining a central radial linear magnification region surrounded by a constrained radial non-linear periphery. Their system was deployed on a video see-through HMD, the system employed a gaze-contingent activation mechanism, enabling magnification when the operator's sight converged on the operative field ($d < 50cm$ and $\alpha < 15^\circ$). A within-subjects study with five surgeons demonstrated that the linear design yielded faster performance initially associated with familiarity with optical loupes; the hybrid design proved superior following adaptation, highlighting the role of learning effects in surgical VML design.

Kluge et al. [32] introduced a VML for the Oculus Rift featuring a circular lens rendered as a disk with a distinct orange-bordered ring. Lenses are spawned via a radial menu and, once placed in 3D space, can be grabbed and manipulated with six degrees of freedom (6DoF), resized through two-handed interaction along the ring border, and parametrised via UI controls embedded directly on the lens. Multiple lenses can be overlapped to produce temporary composite magnification effects or snapped together permanently. For out-of-reach lenses, a semi-transparent proxy appears in front of the user to enable remote manipulation. The system was evaluated with over a hundred participants; however, no controlled experiment or quantitative performance measures were reported. Notably, the coupling between head orientation and lens selection introduces instability under involuntary movement, a limitation that remains unaddressed.

Qian et al. [47] proposed an AR Loupe, a hybrid system coupling a Galilean magnification lens mounted in front of an optical see-through HMD (Magic Leap One) to render AR guidance directly within the magnified field of view. Unlike fully virtual approaches, the design produces two concentric circular regions: an inner magnified zone and an outer annular ring used to recover information occluded by the loupe hardware. A four-stage rendering pipeline manages scene update, camera rendering, magnification, and post-rendering warping, with per-user calibration to align virtual content across both displays. A user study with nine computer science students demonstrated sub-millimetre guidance accuracy ($d = 0.82$), significantly outperforming a baseline AR condition without magnification. However, the small sample of non-domain participants limits the generalisability of these findings to professional clinical

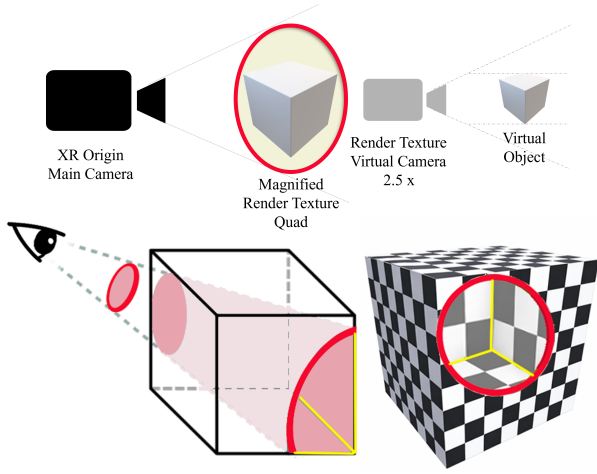


Figure 2: **Conceptual illustration of VMLs.** The rendering pipeline (top): a secondary virtual camera captures the scene at 2.5 $\times$ magnification into a render texture. *Bottom Left*: a volumetric region constrains and guides user motion along a preferred direction within 3D space. *Bottom Right*: application of the VML within a cubic workspace, where the circular cross-section represents the active lens area that modulates visual feedback within the volume.

settings.

Ilo et al. [27] proposed a gaze-controlled VML using a video see-through HMD (Varjo XR-1) that simulates optical magnification by cropping and rescaling the headset’s camera feed to fill the user’s field of view, enabling hands-free variable zoom between 1 $\times$ and 4 $\times$. The interaction design prioritised a hands-free approach suited to the surgical context, realised through two gaze-based techniques: a discrete dwell-activated radial menu offering four preset magnification levels, and a continuous wink-controlled virtual slider for finer adjustment. Voice and gesture input were explicitly excluded to avoid disrupting surgical communication and workflow. A feasibility study with six male surgeons and surgical residents, most with no prior AR/VR experience, found that while participants generally completed the suturing tasks, overall usability was rated poorly and task completion time increased relative to physical loupes. These shortcomings were attributed to the absence of stereo vision, lack of colour passthrough, degraded resolution, device weight, and constrained downward viewing angle.

Schubert et al. [51] introduced a real-time AR magnification system (HoloLens 2) built around a rectangular window metaphor, departing from the circular lens prevalent in prior work. The window is defined through one of three hand-based techniques: unimanually between index finger and thumb, bimanually between both hands, or as a persistent situated rectangle resized via pinch-and-drag at its corners. Magnification scale is controlled through five alternatives: scale-by-size, scale-by-distance, scale-by-voice, scale-by-device, and scale-by-slider. A within-subjects study with 20 participants reported subjective and qualitative differences across conditions; however, no quantitative performance measures such as task completion time or spatial accuracy were reported.

Kern et al. [31] proposed a VML implemented as an augmented mirror rendered as a flat planar surface in VR, where a virtual camera captures the mirror reflection, crops it by a factor of three, and interpolates it back onto the mirror plane via a surface shader, toggled on demand via a VR controller button. A user study ($N = 14$) on a rod alignment task demonstrated that the magnified mirror outperformed both the standard mirror and no-mirror conditions, achieving the highest rotational accuracy ($\alpha = 0.08^\circ$) and closest

distance ($d = 0.61$ cm), though participants required an acclimation period to fully benefit from the magnified view.

Zeng et al. [67] implemented VMLs using video-passthrough AR (Varjo XR-1 and XR-3) to provide gaze-driven digital magnification for microsurgical suturing. Magnification is controlled through one of two gaze-based zoom methods: discrete zoom via a floating pop-up menu displayed in the AR scene, or continuous zoom via eye winking, both driven by the headset’s integrated eye tracker. The system was evaluated across two phases with six participants performing skin and anastomotic suturing simulations with progressively finer suture material. No strong preference between zoom methods was observed, though discrete zoom was perceived as easier to use, and hardware improvements between phases yielded significantly higher System Usability Scale (SUS) scores. Key unresolved limitations include degraded depth perception and reduced suture visibility arising from the constraints of digital magnification.

While prior systems have established the technical viability of virtual magnification in procedural contexts, the field lacks a systematic investigation of the core design dimensions that determine practical utility. In particular, the spatial coupling between the magnification lens and the user’s viewpoint, specifically whether the lens is anchored in the world or moves with the user’s head, has not been treated as a variable worthy of direct empirical comparison. Evaluations have been further limited by small or non-practitioner samples, abstract task proxies, and a reliance on subjective measures that do not capture clinically relevant performance.

3 METHODOLOGY

3.1 Co-Design Process

The co-design methodology [66, 23, 45] involved two expert dentists (with 8 and 15 years of clinical experience) over a six-month period. The process comprised eight iterative design-review sessions structured across three phases.

Phase 1: Elicitation: Semi-structured interviews with both dentists identified core clinical requirements, including preferred magnification levels (2-3 $\times$), optimal lens placement relative to the oral cavity, and the critical importance of preserving bimanual workflow without hand-based interaction for loupe control.

Phase 2: Prototyping: Three candidate designs were prototyped (see Fig. 3): a head-coupled loupe, a world-fixed loupe, and a hand-anchored loupe. The hand-anchored variant was eliminated following internal review, as it interfered with the bimanual instrument handling required in dental procedures. The remaining two designs were iteratively refined based on dentist feedback regarding spatial placement, magnification behaviour, and motion coupling.

Phase 3: Validation: Both dentists performed representative tasks in the VR environment and provided structured feedback on the final loupe parameters, leading to the specifications reported in Tab. 2.

3.2 Implementation

Based on the co-design process, three experimental conditions were defined. All conditions (see Fig. 3) were implemented in Unity (version 6000.0.60f1) using the XR Origin framework. The virtual environment, patient model, interaction tools, and rendering pipeline were held constant across conditions, with differences limited to the presence and behaviour of the virtual loupe. This ensured that observed effects could be attributed to the magnification strategy (see Fig. 2) rather than environmental variation. The technical specifications of both VML implementations are summarized in Tab. 2.

3.2.1 No Loupe Baseline

A baseline design with no visual magnification. The NL condition served as a non-magnified reference baseline against which

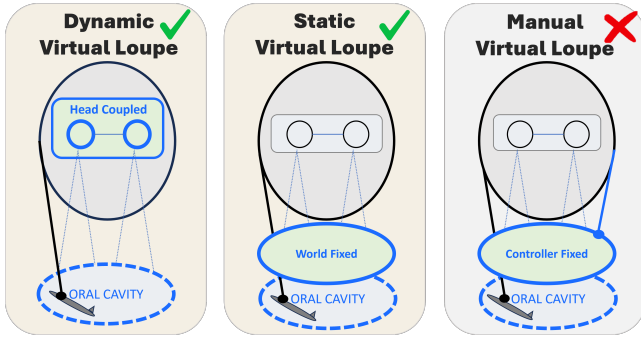


Figure 3: **The three candidate loupe designs explored during the co-design process:** (left) Dynamic Virtual Loupe (DVL), in which the loupe is head-coupled and the virtual field of view moves with the dentist's head; (center) Static Virtual Loupe (SVL), in which the loupe is anchored in world space near the patient's oral cavity; (right) Manual Virtual Loupe (MVL), in which the loupe position is controlled by hand, occupying one hand and precluding bimanual instrument use.

the perceptual and motor effects of both loupe configurations could be isolated and quantified. Participants performed identical task trials within the same virtual dental environment using the native stereo rendering pipeline of the Quest 2, with no auxiliary cameras, render textures, or overlay geometry introduced into the scene. Visual inspection of the operative field relied entirely on natural head movement and operator-controlled proximity, replicating the unaided visual conditions characteristic of pre-loupe clinical practice. This condition establishes the performance baseline necessary to attribute observed differences in accuracy, task completion, and postural metrics to the presence and configuration of virtual magnification.

3.2.2 Dynamic Virtual Loupe (DVL)

The DVL is a head-coupled stereoscopic magnification system employing two eye-aligned virtual lenses that follow the user's head orientation (see Fig. 4). It was designed to replicate the optical and behavioural properties of head-mounted surgical magnification loupes within an immersive VR environment. The system employs two quad primitives, one per eye, rendered via dedicated auxiliary cameras outputting to RenderTexture targets. Each auxiliary camera operates at a constrained field of view of 24° , yielding an optical magnification factor of $2.5\times$, derived from the ratio $M = \text{FOV}_{\text{main}} / \text{FOV}_{\text{aux}}$, where the Quest 2 monocular horizontal FOV is approximately 60° .

The lens quads and their corresponding auxiliary render texture cameras (see Fig. 2) are rigidly parented to the headset camera within the XR Origin hierarchy, ensuring that the magnified

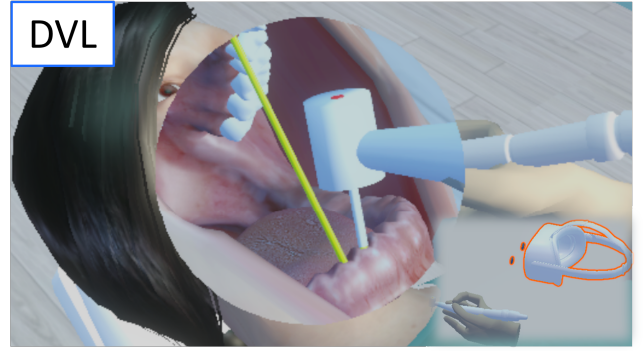


Figure 4: DVL loupes are rigidly coupled to the headset, remaining continuously visible within the user's field of view regardless of head orientation.

viewport remains continuously present within the user's field of view throughout the task. The inter-lens separation is configurable across three discrete settings spanning the Quest 2's supported inter-pupillary distance range of 56-70 mm, and was adjusted per participant prior to each session to ensure accurate binocular alignment and visual comfort. To mitigate perceptual disturbance introduced by the persistent lens overlay, a parametric background blur was applied to the peripheral region surrounding each quad, attenuating visual clutter without compromising the fidelity of the magnified region. This rigid head-coupling reproduces the fundamental constraint of physical loupes: magnification direction is inseparable from head orientation, requiring the operator to actively stabilise posture and direct gaze toward the operative field to extract meaningful visual information from the magnified view.

3.2.3 Static Virtual Loupe (SVL)

The SVL is a world-fixed virtual magnification lens positioned relative to the patient's oral cavity, remaining stationary regardless of head movement (Fig. 5). It was designed to investigate whether spatially decoupling the magnification lens from head motion yields measurable benefits in procedural performance and postural behaviour. In contrast to the DVL, the SVL lens is instantiated as a world-fixed object within the virtual environment, anchored at a fixed position 8 cm from the target site within the patient's oral cavity. This placement was informed by expert clinician input gathered during the co-design process, reflecting ecologically valid working distances in operative dentistry.

The optical implementation mirrors that of the DVL: an auxiliary camera with a 24° FOV renders to a RenderTexture displayed on a quad primitive, producing equivalent $2.5\times$ magnification. However, neither the quad nor the auxiliary camera maintains any kinematic relationship with the headset transform; their world-space coordinates remain invariant across the duration of each trial. Consequently, the magnified view is not continuously available to the user but must be actively sought by aligning the operator's viewpoint with the fixed lens aperture. This design imposes a spatial constraint on magnification access, effectively penalising erratic head movement and incentivizing the adoption of a stable, task-optimal working posture. The absence of a persistent overlay eliminates the need for background blur, as the lens occupies a bounded region of the scene rather than the user's visual field.

Table 2: Technical specifications of the two VML implementations. Both provide identical magnification factor and auxiliary camera field of view; they differ in spatial coupling strategy.

Parameter	DVL	SVL
Magnification factor	$2.5\times$	$2.5\times$
Auxiliary camera FOV	24°	24°
Lens diameter	20 mm	300 mm
Lens distance from headset camera	Head-fixed 500 mm	N/A
Lens distance from target	N/A	150 mm
Motion coupling	Head-fixed Yes	World-fixed No
Stereoscopic rendering	(lens per-eye)	(single lens)

4 DESIGN OF EXPERIMENT

The study employed a within-subjects design in which each participant completed task trials across all three experimental conditions in Latin-square order. This design controls for inter-subject variability in motor skill and prior dental experience, ensuring that

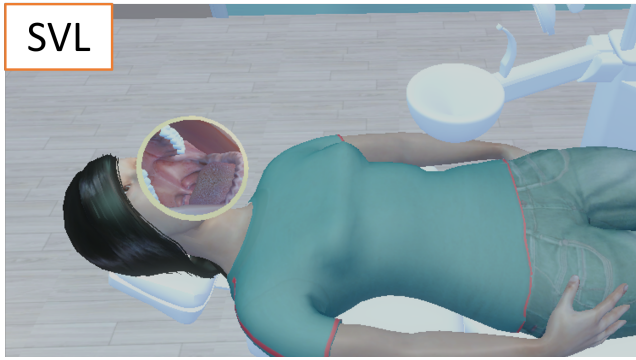


Figure 5: **SVL** loupe is anchored in world space adjacent to the patient's oral cavity, requiring the operator to actively align their view-point with the fixed aperture to access the magnified view.

observed differences in performance and perceived workload are attributable to the magnification configuration rather than individual ability.

4.1 Research Questions and Hypotheses

Based on the identified gap, this study investigates the following research questions:

RQ₁: Does the interaction design of a VMLs differentially affect procedural performance and perceived cognitive load in immersive dental training?

RQ₂: Which VML design do users prefer, and what factors drive (perceptual, motor, or cognitive) that preference?

Based on cognitive load theory and prior work on magnification in precision tasks, the following hypotheses were formulated:

- **H₁:** **DVL will yield superior performance compared to SVL and NL.** Continuous availability of magnified visual feedback is expected to produce significantly lower positional error (mm), rotational error (°), and task completion time (s) relative to both SVL and NL.
- **H₂:** **Both VML designs will reduce perceived workload compared to the NL baseline.** Optical magnification reduces visual search demands and supports finer spatial judgment during precision tool manipulation.
- **H₃:** **DVL will produce lower perceived temporal demand than SVL.** Head-coupled magnification eliminates the spatial realignment overhead inherent to accessing a world-fixed aperture, reflected in a significantly lower NASA-TLX temporal demand sub scale score.
- **H₄:** **Both VML designs will reduce perceived workload compared to NL.** Optical magnification reduces visual search demands and supports finer spatial judgment, expected to yield a statistically significant reduction in NASA-TLX total score for both DVL and SVL relative to NL.
- **H₅:** **DVL will emerge as the preferred design.** The intuitive correspondence between head orientation and magnification access that characterizes physical loupe use in clinical practice is expected to drive a statistically significant preference toward DVL as measured by ranked choice response.

4.2 Participants

The experiment was approved by our institutional review board. A power analysis conducted prior to recruitment (G*Power 3.1; repeated-measures ANOVA, within-subjects; $f = 0.30$, $\alpha = .05$,



Figure 6: **Experimental setup in a simulated dental room.** (Top-left) Participant seated in a dental chair wearing a Meta Quest 2 HMD, interacting with a dental phantom. (Top-right) Close-up of the phantom and handheld controller used for spatial alignment with the real-world counterpart. (Bottom-left) Full-room view replicating a clinical dental environment, with the participant using a foot pedal for task confirmation. (Bottom-right) 3D-printed drill proxy controller, used to simulate instrument manipulation in the virtual environment.

power = .80, 3 conditions, correlation among repeated measures = .50) indicated a minimum sample of 22 participants, ensuring sufficient statistical power. Twenty-five dental students voluntarily participated (12♀, 13♂; mean age = 26 years, range 22-35, $SD = 3.78$). All were right-handed and enrolled in semesters 3-10 ($M = 7$ th semester, $SD = 2.36$) at a university dental program. Participants reported modest XR familiarity ($M = 3.0/7$, $SD = 1.93$) and gaming experience ($M = 3.0/7$, $SD = 1.74$) on 7-point Likert scales (1 low - 7 high). All participants reported using physical magnification loupes before but none reported prior experience with virtual ones.

4.3 Setup

Following the validated approach of simulating AR within VR [36, 48] we adopted the same methodology for our MR condition, which occupies the same reality-virtuality continuum [41]. The experiment was conducted within an operational clinical dental room at an anonymized university hospital, ensuring that the environmental conditions spatial layout, ambient lighting, chair configuration, and instrumentation were representative of authentic clinical practice. The physical setup was established in direct collaboration with expert dentists throughout the co-design process, resulting in a configuration that closely mirrors a real-world dental workstation while maintaining full experimental control over task parameters, condition ordering, and interaction variables.

The setup comprised the following components (Fig. 6):

- A Oculus Quest 2 HMD (Meta, CA, USA) connected via Oculus Link to a laptop (NVIDIA RTX 3060, Intel i7-12700H) running the Unity application;
- A dental phantom as described in literature [57, 37] mounted on a standard patient chair, providing realistic spatial relationships between dentist, patient, and instruments;

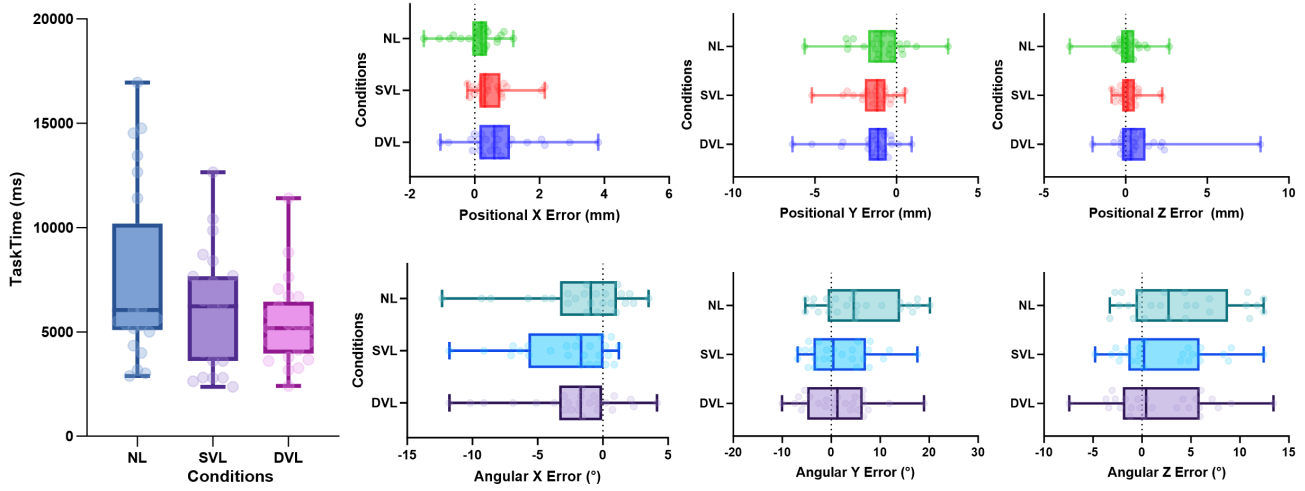


Figure 7: **Objective Results: Task Time, Positional and Angular Errors (X, Y, Z Axis).** Box plots with scatter plots display distributions of task time (ms), positional error (mm), and angular error ($^{\circ}$) across DVL, SVL, and NL conditions ($n = 25$). No significant differences were found across conditions for any objective measures.

- A foot pedal (Xiatiaosann, Guangzhou, China), similar function and placement of the clinical dental practice to confirm drill positioning;
- Oculus Quest 2 controllers (Meta, CA, USA) ; one used as proxy for the virtual drill tool, and one used to track the virtual content with the real one.

4.4 Task Design

Each task trial required participants to position a virtual dental drill at a specified target location and orientation within the patient's oral cavity (Fig. 6). Target positions were defined as 3D coordinates with associated orientation vectors, representing clinically plausible drill placements for common dental procedures. The targets are represented with yellow cylinders (Fig. 6) and participants were informed that the intersection of the cylinder and the patient's gingiva is the target position. The target orientation, instead, is the angulation of the cylinder. On top of the drill tool, represented as a red circle, an additional visual aid helps participants position the tool orientation.

Upon achieving satisfactory drill tool positioning and alignment, participants confirmed the trial by pressing the foot pedal, which logged the position and orientation error and advanced to the subsequent trial. Targets were identical across all three designs and presented in the same fixed sequence within each design block. This ensured that design comparisons were not confounded by target difficulty variation.

4.5 Procedure

Participants first received a standardized oral briefing covering the study aims, task instructions, and each design's interaction mechanics. Informed consent was obtained, and participants were informed they could withdraw at any time.

After the introduction, participants were seated in the dental chair and wore the VR headset. A training phase consisting of five tool positioning trials as practice per condition allowed participants to familiarize with the interaction mechanics and magnification behaviors before data collection.

During the main experiment, condition order was counterbalanced across participants using a balanced Latin square design to control for order and carryover effects. Each condition block consisted of ten tool positioning trials, yielding 30 trials per participant (750 total across the study).

Upon completing all three condition blocks, participants removed the headset and completed the NASA-TLX questionnaire for each condition. An oral interview followed, in which the experimenter collected subjective impressions, condition preferences, and suggestions for improvement. The entire user study took approximately 30-40 minutes.

4.6 Metrics

4.6.1 Objective Measures

- **Positional error:** distance between the drill tool tip and the target position (mm), decomposed into X (horizontal), Y (vertical), and Z (frontal) axis components for analysis.
- **Rotational error:** Angular deviation between the drill orientation and the target orientation (degrees), for each X, Y, and Z axes.
- **Task time:** Duration between two tool-positioning trials, counted between consecutive foot pedal presses (seconds).

4.6.2 Subjective Measures

- **NASA-TLX:** Six dimensions (Mental Demand, Physical Demand, Temporal Demand, Effort, Perceived Performance, Frustration) rated on 7-point Likert scales (1 = low, 7 = high). Note that for Perceived Performance, lower scores indicate better self-assessed performance.
- **Preference:** overall preferred condition as declared by the user.
- **User opinion:** Semi-structured interview with Open-ended questions on comfort, usability, and suggestions.

4.7 Statistical Analysis

For each participant and condition, positional and orientational errors were aggregated by computing the mean across all trials, yielding a single representative score per participant per condition prior to inferential testing. Statistical tests were selected based on the measurement scale and distributional properties of each dependent variable.

Since the study followed a within-subjects design, repeated-measures analyses were employed throughout. Normality was assessed using the Shapiro-Wilk test. As violations of normality

were identified across both objective performance measures (task completion time, positional error, and angular error) and subjective measures (NASA-TLX subscales), Friedman tests were used for all inferential analyses. Post-hoc pairwise comparisons were conducted using Dunn's test with Bonferroni correction.

User preference was collected as a single first-choice selection per participant and analysed using a chi-square goodness-of-fit test against a uniform distribution, testing whether the observed preference frequencies deviated significantly from chance. The significance threshold was set at $\alpha = .05$ for all tests.

5 RESULTS

5.1 Objective Results

Descriptive statistics are summarized in Tab. 3 and inferential results are illustrated in Fig. 7. No significant differences across conditions for task completion time ($\chi^2(2) = 1.280$, $p = .527$, $W = .03$), positional error on the X-axis ($\chi^2(2) = 4.160$, $p = .125$, $W = .08$), Y-axis ($\chi^2(2) = 1.040$, $p = .595$, $W = .02$), or Z-axis ($\chi^2(2) = 1.520$, $p = .468$, $W = .03$), nor for angular error on the X-axis ($\chi^2(2) = 1.680$, $p = .432$, $W = .03$), Y-axis ($\chi^2(2) = 0.320$, $p = .852$, $W = .01$), Z-axis ($\chi^2(2) = 0.240$, $p = .887$, $W = .00$), or overall angular error magnitude ($\chi^2(2) = 2.67$, $p = .263$, $W = .05$). These results indicate that neither VML design conferred a measurable advantage in spatial accuracy or task efficiency relative to the NL baseline.

5.2 Subjective Results

Subjective results are illustrated in Fig. 8. In contrast to the objective measures, both VML designs produced significant reductions in perceived cognitive workload across several NASA-TLX subscales relative to NL.

Mental demand was significantly lower for both DVL ($M = 3.2$, $SD = 1.4$; $p = .042$) and SVL ($M = 3.0$, $SD = 1.3$; $p = .001$) compared to NL ($M = 4.5$, $SD = 1.6$), with no significant difference between the two VML designs ($p = .438$). SVL produced significantly lower physical demand than NL ($M = 2.9$ vs $M = 3.9$; $p = .018$), while DVL did not differ significantly from either condition. The largest effect emerged for temporal demand ($\chi^2(2) = 18.45$, $p < .001$, $W = .37$), with DVL ($M = 2.3$, $SD = 1.1$) rated significantly lower than SVL ($M = 3.2$, $SD = 1.3$; $p = .046$), which in turn was rated significantly lower than NL ($M = 4.5$, $SD = 1.4$; $p < .001$). SVL also required significantly less effort than NL ($M = 3.1$ vs $M = 4.2$; $p = .012$), while DVL fell between the two without reaching significance. DVL was rated most favourably for perceived performance ($M = 2.1$, $SD = 1.0$), followed by SVL ($M = 3.3$, $SD = 1.2$; $p = .004$) and NL ($M = 4.4$, $SD = 1.5$; $p < .001$). Frustration showed no significant effect across conditions (all $p > .05$).

Taken together, these findings indicate that both VML designs consistently reduced perceived cognitive workload relative to NL, with DVL showing the broadest benefit across subscales.

5.3 User Preference

DVL was the most preferred design, selected by 15 of 25 participants, followed by SVL with nine selections, and NL with only

one. A chi-square goodness-of-fit test confirmed that the distribution of first-choice preferences deviated significantly from chance ($\chi^2(2) = 11.84$, $p = .003$), indicating a clear overall preference toward VML-assisted conditions over the NL baseline.

6 DISCUSSION

The central finding of this study is that both VML designs preserved task performance while significantly reducing perceived cognitive workload. Across all three conditions, participants achieved statistically equivalent task completion time, positional error, and angular error, thereby failing to confirm H_1 . Both VML designs significantly reduced perceived workload relative to NL, confirming H_2 , with DVL producing the lowest temporal demand, confirming H_3 , and the highest perceived performance across all conditions. In the preference rankings, 15 of 25 participants chose DVL, nine chose SVL, and only one preferred NL, confirming H_5 .

Similarly to our results, conventional studies report varying effects of optical MLs on objective outcomes depending on the clinical task [21, 16], and the absence of significant differences in objective performance here should therefore not be interpreted as an absence of benefit. The virtual task environment likely reduced effective precision demands relative to real clinical settings: the immersive environment provided a target marker previously shown to be a preferred navigational aid among dental students, achieving higher placement precision compared to unaided task completion [18]. Furthermore, it lacked clinically relevant confounds such as patient movement, intraoral access constraints, and haptic feedback.

Participants were dental students with multiple semesters of clinical training, experienced enough to achieve the required precision for this task without magnification assistance. Together, these factors suggest that the task fell within the compensable range of unaided performance for this population: participants could reach the same precision ceiling regardless of condition, but the cognitive and perceptual resources required to do so varied substantially across conditions. Under the NL condition, participants compensated for the absence of magnification by leaning closer, stabilising hand posture, and relying on practised binocular depth cues to maintain target accuracy. Such compensatory strategies are well-documented in the motor learning literature [30]. They are effective but they are costly: each draws from the same finite attentional pool [63] that the practitioner simultaneously requires for clinical judgment and patient monitoring.

This has direct implications for how augmented visual interfaces should be evaluated. The conventional approach treats objective performance improvement as the primary criterion for benefit [20]. For skilled populations performing tasks within their compensable range, however, this criterion systematically underestimates the value of well-designed visual aids. A more appropriate question is: at what cost was equivalent performance achieved? The NASA-TLX results address this directly: both VML designs significantly reduced mental and temporal demand relative to NL, indicating that the magnified view offloaded spatial interpretation from the practitioner. In domains such as dentistry, where practitioners must simultaneously manage fine motor execution, clinical decision-making, and patient communication [2], reducing the perceptual cost of motor tasks preserves attentional capacity for the higher-order clinical judgments that ultimately determine outcome quality and patient safety.

This finding aligns with a broader principle in augmented interface design [6]: the most significant effect of a well-designed visual aid is often not an improvement in peak performance but a reduction in the attentional budget required to sustain a given performance level [34]. For virtual magnification loupes specifically, this reframing suggests that cognitive load reduction should be treated as a primary design and evaluation criterion alongside task accuracy.

Table 3: Descriptive statistics for objective performance measures across conditions ($M \pm SD$).

Metric	DVL	SVL	NL
Task time (ms)	5401 $\pm$ 1951	6014 $\pm$ 2728	7701 $\pm$ 4047
Positional error (mm)	2.19 $\pm$ 1.09	2.17 $\pm$ 1.40	2.72 $\pm$ 1.60
Angular error ($^\circ$)	7.57 $\pm$ 3.25	8.72 $\pm$ 4.78	10.68 $\pm$ 7.02

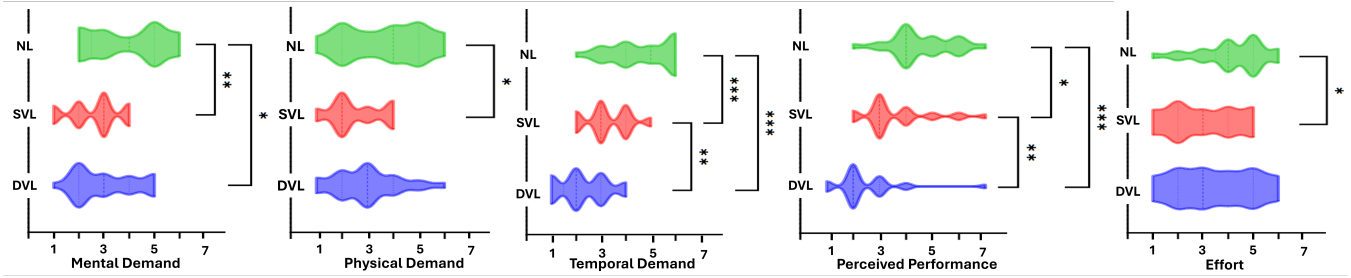


Figure 8: **Subjective results: NASA-TLX.** Violin plots display distributions of subjective workload ratings for Mental Demand, Physical Demand, Temporal Demand, Perceived Performance, Effort ($n = 25$). Significant pairwise differences are indicated (* $p < .05$, ** $p < .01$, *** $p < .001$). DVL produced the lowest scores on Temporal Demand and Perceived Performance, while both VML conditions significantly reduced Mental Demand relative to NL.

6.1 Dynamic vs. Static Virtual Loupe

Although the DVL and SVL shared similar optical parameters (Tab. 2), they produced noticeably different subjective measure outcomes. These differences can therefore be understood as the consequences of the design, which is the spatial relationship between the loupe and the practitioner.

The DVL produced the lowest temporal demand ratings across all comparisons. This difference is likely a direct consequence of how each design manages attentional continuity. Since the head-coupled loupe remains persistently within the operator's field of view, the magnified image is available without requiring any spatial reorientation or visual search. The operator simply directs their head-gaze toward the task area, and the magnified view is already present. In contrast, accessing the SVL requires the operator to align their viewpoint with the fixed aperture, a spatial and cognitive overhead that accumulates across repeated task cycles.

The SVL, however, was rated lowest on physical demand. The world-fixed lens encouraged participants to adopt and maintain a stable working posture aligned with the task site. It also allows other tasks to remain unmagnified, such as tool selection and handling, reducing the frequency of postural adjustments. The DVL permits more dynamic movement, which sustains magnification availability independent of head movement but at the cost of slightly greater physical engagement. The two designs thus impose different postural disciplines: the DVL tolerates movement without constraining it, while the SVL rewards stillness.

The DVL also produced significantly higher perceived performance ratings, a consequential finding from a training perspective. The continuously available magnified view did not merely reduce effort; it increased participants' confidence in the precision of their own actions. By contrast, the SVL left participants less certain of their performance, likely because the intermittent nature of magnification access introduced ambiguity about whether the final drill position had been verified with sufficient visual resolution prior to confirmation. In an educational context, this confidence effect may support the development of loupe-assisted clinical skills; however, perceived confidence must be interpreted with caution, as it does not necessarily reflect actual performance accuracy [24, 1].

The complementary profiles of the DVL and SVL converge on a clear design principle: matching the magnification strategy to task demands is likely to yield greater benefits than committing to a single fixed configuration. This motivates a design direction that follows naturally from the findings but was not evaluated in the present study: a system in which the practitioner can select among loupe configurations at runtime. Such a framework would allow operators to switch between head-coupled magnification during precision-critical phases, world-fixed magnification during sustained monitoring phases where postural stability matters more than instantaneous access, and hand-anchored magnification in contexts

where a free hand permits direct spatial control over the magnified region. The latter variant was discarded during co-design due to the bimanual constraints of dentistry (see Fig. 3), but remains viable in other domains or settings. The foot-pedal, voice commands, or gaze-based selection could serve as mode-switching inputs, preserving a bimanual workflow while enabling runtime reconfiguration. The critical principle is that magnification strategy should be a practitioner-controlled parameter rather than a fixed system property, consistent with the broader trend toward adaptive interfaces in mixed reality [26].

6.2 Clinical Dental Perspective

Dentistry imposes a unique combination of constraints that few other precision domains share simultaneously: the intraoral workspace is anatomically irregular, poorly illuminated, partially occluded by soft tissue and saliva, and accessible only through the mouth opening, limiting both instrument angles and sight-lines [29, 28]. These constraints push many clinical tasks—caries detection, crown preparation, endodontic access—to the edge of unaided visual resolution [17, 7, 46], which is why physical loupes have become ubiquitous. A VML that performs well under these conditions is likely to generalise to less demanding precision tasks.

The workload reduction observed here carries a concrete clinical implication: extended cognitive endurance. During multi-hour restorative sessions, accumulated cognitive fatigue degrades decision quality over time; any tool that lowers per-task attentional cost effectively extends the window of reliable performance. The DVL's advantage in temporal demand is particularly relevant to procedures such as composite layering or crown preparation, where the clinician must repeatedly verify sub-millimetre spatial relationships between the instrument tip and the prepared margin. In these workflows, momentarily losing the magnified view risks disrupting the practised motor rhythm on which experienced clinicians rely [30].

Modern loupes already improve posture through built-in declination angles, yet the SVL may confer an additional ergonomic benefit [5, 61]. By anchoring the view at the operative site and reducing the need for continuous head tracking, the SVL further minimises postural adjustments during procedures. Although the short-trial protocol of the present study could not capture these long-term effects, the finding suggests that the optimal magnification strategy may be procedure-dependent: DVL for short, dynamic tasks such as multi-tooth diagnostics, and SVL for prolonged single-tooth procedures where sustained posture outweighs instantaneous access.

The co-design process was pivotal in surfacing these trade-offs. Several essential design decisions emerged from clinical observation rather than technology-driven reasoning: the $2.5\times$ magnification factor, matching the most commonly prescribed clinical loupes [43]; the 15 cm lens-to-target distance, reflecting realistic working ranges; and, above all, the foot-pedal activation, preserv-

ing the bimanual mirror-and-instrument workflow fundamental to operative dentistry. The early elimination of the hand-anchored variant illustrates the point: although technically feasible and potentially useful where one hand is free, a loupe that occupies a hand is clinically unacceptable in a discipline where both hands are continuously engaged with instruments and retraction. Such domain-specific constraints remain invisible without sustained clinical exposure, underscoring co-design as more than a methodological formality.

Finally, while VMLs offer clear dynamic design advantages over physical optics, practical adoption depends on VR infrastructure that remains costly and scarce in most dental training settings. A systematic review of mixed reality in the operating room identifies tracking precision and the absence of practitioner-driven interface customisation as the primary barriers to broader surgical uptake [38]. As these barriers diminish, the configurability demonstrated here could position virtual loupes as a flexible, cost-effective complement to conventional magnification.

6.3 Design Implications Beyond Dentistry

The findings generalize to any domain in which practitioners have to perform fine spatial manipulations within constrained workspaces. Recent findings from industrial AP assembly guidance confirms that AR-guided operators matched or exceeded conventional performance while reporting significantly lower cognitive load [14]. In each case, The empirical results reported here provide a principled basis for that decision. Tasks requiring sustained postural stability over extended durations should favor SVL world-fixed alternatives. We recommend that future VML implementations in other domains follow a similar practitioner-in-the-loop design process rather than adopting parameters from prior systems without contextual validation.

6.4 Limitations and Future works

Several limitations of this study that should be addressed by the future research.

This study employed a VR-based simulation of a mixed-reality environment for two reasons: first, VR allowed us to directly compare different loupe techniques under controlled conditions; second, the simulated MR setting bypassed the tracking limitations inherent in current optical see-through displays. However, future work should implement these design principles in true mixed-reality setups to evaluate their clinical impact during realistic procedures on dental phantoms or patients.

The ten-trial task block length per condition limited the adaptation time; longitudinal studies with extended practice could reveal whether the SVL's initial access overhead diminishes with experience, potentially narrowing the subjective gap with DVL. The results also may not generalize straightforwardly to all clinical tasks in dentistry. Future work should also investigate different tasks with exceed unaided visual acuity by requiring sub-millimeter positioning tolerances or operating with reduced illumination conditions.

The resolution and field of view of the Meta Quest 2 limited the maximum fidelity of the magnified view. Next-gen headsets with higher-resolutions may amplify or alter the performance differences observed here. Similarly, the absence of gaze behavior or head motion trajectories data limited our ability to validate the postural and attentional mechanisms, future studies should incorporate these measures to provide converging physiological evidence. Nevertheless, the VR-based evaluation paradigm presented in this work is inherently versatile, because the VMLs parameters are defined in software, the same experimental framework can be readily adapted to different headsets, magnification configurations, and task domains without requiring physical tooling or optical hardware changes.

The participants were dental students rather than practicing clinicians; expert operators with more automatized motor routines may respond differently to magnification assistance. Finally, the combined VML design systems that switch between head-coupled and world-fixed modes based on task phase (e.g., coarse cavity preparation vs. fine margin verification) or user configurable systems should be further explored, leveraging the complementary strengths identified in this study.

6.5 Takeaways

- The co-design with experts was key, bringing empirical decisions and critical perspectives not possible from technology-driven reasoning alone: the magnification factor, lens placement, foot-pedal activation, and eliminated the hand-anchored loupe.
- All VML variants demonstrated to reduce dentists' cognitive workload, with the benefit to allocate more attention to the medical procedures.
- DVL and SVL have different interaction strengths, so they better work synergistically according to the task: head-coupled DVL favors attentional continuity, temporal demand efficiency, and perceived performance, while world-fixed SVL favors physical comfort and postural stability.
- Magnification strategy should be a user-configurable runtime parameter, not a fixed system property, enabling practitioners to match loupe behavior to procedural phase, task demands, and individual preference.
- With DVL participants rated their own performance highest, leading the risk of overconfidence. Training must be calibrated against objective measures in clinical setting.

7 CONCLUSIONS

Both virtual magnifying loupes preserved task accuracy and speed while significantly reducing cognitive workload. For skilled operators already performing at ceiling, the value of a visual aid lies not in what it improves but in what it spares: attentional resources freed from spatial verification become available for clinical reasoning and error monitoring. These results establish spatial coupling as a consequential design variable for precision tasks in mixed reality, motivating future systems that treat loupe configuration as user selectable rather than fixed.

SUPPLEMENTARY MATERIALS

An anonymous video illustrating the study conditions is provided as supplementary material. Data availability: All anonymized experimental data supporting the findings of this study are provided as supplementary material to enable replication and secondary analysis. Source availability: Both VML implementations, including source code and graphical assets, released as open-source to foster future research in virtual magnification design.

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